A. .A To show x is injective, we consider two cases:
Case 1 When $n = 1$, x is never injective, since we have many ways to map to a single real number.
Case 2 When $n \geq 2$, x is injective if and only if the columns of A are linearly independent.
To show x is surjective, we consider two cases:
Case 1 $n \geq 3$. In this case, x is never surjective because its image has at most dimension 2.
Case 2 $n = 2$. In this case, we can construct any (a, b) by letting $(u, v) = A^{-1}(a, b)$, and x is surjective.
Case 3 $n = 1$. In this case, x is surjective when $A \neq 0$.

.B We compute

$$\frac{\partial x}{\partial u} = \frac{\partial}{\partial u} A \begin{bmatrix} u \\ v \end{bmatrix} = A \begin{bmatrix} \frac{\partial}{\partial u} u \\ \frac{\partial}{\partial u} v \end{bmatrix} = A \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \frac{\partial x}{\partial v} = \frac{\partial}{\partial v} A \begin{bmatrix} u \\ v \end{bmatrix} = A \begin{bmatrix} \frac{\partial}{\partial v} u \\ \frac{\partial}{\partial v} v \end{bmatrix} = A \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

We again have cases. When the rank of A is less than 2, u and v are linearly dependent and x is not a regular surface. When rank $A = 2$, then x is a regular surface.

B. .A Let $x(u_1, v_1) = x(u_2, v_2)$. Then, $({u_1}^3, v_1) = ({u_2}^3, v_2)$. The function $f(x) = x^3$ is injective, and $v_1 = v_2$. Therefore, x is injective.
Let (a, b) be a point in $\mathbb{R}^2$. Let $u = a^{1/3}$ and $v = b$. We then have $x(u, v) = (a, b)$. Thus, x is surjective.

.B We compute

$$\frac{\partial x}{\partial u} = (2u^3, 0), \frac{\partial v}{\partial u} = (0, 1).$$

Anytime $u = 0, v = c$, $\frac{\partial x}{\partial u} = (0, 0)$, and therefore is not linearly independent with $\frac{\partial x}{\partial v}$. Therefore it is not a regular surface.

C. .A SKETCH THIS The x_1 and x_2 components of x parameterize a circle via u in the xy-plane. The x_3 simply takes on the value of $v \in \mathbb{R}$. This surface extends the circle into the z direction, creating a cylinder. We compute $\frac{\partial x}{\partial u} = (-\sin u, \cos u, 0)$ and $\frac{\partial x}{\partial v} = (0, 0, 1)$. By observation, these are linearly independent, and thus the cylinder is a regular surface.

.B SKETCH THIS The shape of is similar to C.A, but the radius of the circle parameterized by u is dependent on v. As such, as the height changes so does the radius of the circle. Thus, this forms a cone with a tip at $(0, 0, 0)$. We compute $\frac{\partial x}{\partial u} = (-v \sin u, v \cos u, 0)$ and $\frac{\partial x}{\partial v} = (\cos u, \sin u, 1)$. Anytime $v = 0$, $\frac{\partial x}{\partial u}$ is the zero vector, and therefore x is not a regular surface.

.C x is not a local parameterization about $p = (1, 0, 1)$ because it is not injective. We can construct a local parameterization by restricting U to $U = (0, \pi) \times (0, \infty)$. Then, we can construct an F defined by $F(x_1, x_2, x_3) = (\arccos(\frac{x_1}{x_3}), x_3)$.

.D There exists no local parameterization about $p = (0, 0, 0)$. Only points $x(u, 0) = (0, 0, 0)$, and this is a line, which is no longer an open subset of $\mathbb{R}^2$.